



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

PROJECT 2

**SECI1143 PROBABILITY & STATISTICAL DATA ANALYSIS
SEMESTER II, SESSION 2022/2023**

SECTION 05

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Group : Group 1

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1.0) Introduction or background

The title of this project is the *Survey on Heart Disease Prediction Among People*. Basically, heart disease refers to several types of heart conditions. For instance, heart disease affects the blood flow to the heart. Decreased blood flow can cause a heart attack. For our information, about 80 percent of heart disease is multifactorial and arises through various combinations of genetic and environmental contributors. Globally, around 110 million men and 80 million women have heart disease. This kills an estimated 9 million people each year. Probably, an estimated 17 million people died from cardiovascular diseases, which is 32 percent of all global deaths. Of these deaths, 85 percent were due to heart attack and stroke. Considerably, heart disease was the biggest killer worldwide before the coronavirus pandemic. Usually, there are many factors of heart disease, such as high blood pressure, diabetes, smoking, obesity, and physical inactivity. So these causes heart disease among the people.

Furthermore, the **purpose of this topic is that we were eager to know about the classification and statistical prediction about heart disease** nowadays. This is because nowadays the cardiovascular risk keeps increasing without stopping. Many of the people face death because of heart disease. So, we want to know actually what is the major factor and classification of people that have heart disease. We were able to get some information about the type of chest pain, blood pressure, cholesterol level, sugar level, exercise time and so on. By getting this information we easily can do classification about the heart disease prediction. We got this information from a dataset in Kaggle by Allena Venkata Sai Aby. Then, we collected some information about classification for heart disease prediction. By this we are able to do data analysis such as correlation, regression and Chi-square test by defining new variables. This data analysis will help us to achieve our purpose of this project.

2.0) Dataset

Dataset - Kaggle link :

<https://www.kaggle.com/datasets/abhishek14398/heart-disease-classification>

Data description :

Variables	Description	Type of variables	Measurement level
age	Age of person	Quantitative	Ratio
sex	Gender of the person	Qualitative	Nominal
chest_pain_type	Type of the chest pain	Qualitative	Nominal
resting_bp	Blood pressure while resting	Quantitative	Ratio
cholesterol	Serum cholesterol	Quantitative	Ratio
fasting_blood_sugar	Blood sugar while fasting	Qualitative	Nominal
restecg	Electrocardiographic while resting (0 : normal, 1: STT abnormality)	Qualitative	Nominal
max_hr	Maximum heart rate achieved	Quantitative	Ratio
exang	Exercise (0 : No , 1: Yes)	Qualitative	Nominal
oldpeak	ST depression induced by exercise relative to rest	Quantitative	Ratio
slope	Slope of the peak exercise ST segment - Value 0: upsloping; Value 1: flat; Value 2: downsloping	Qualitative	Nominal
num_major_vessels	no. of major vessels (0-3) colored by fluoroscopy	Qualitative	Nominal

thal	thalassemia - 0: normal; 1: fixed defect; 2: reversible defect	Qualitative	Nominal
target	0 = no disease; 1 = disease	Qualitative	Nominal

Statistical Test Analysis :

Variables	Objectives	Analysis and Expected Outcome
target and max_hr	To test the difference of two mean of the maximum heart rate achieved by the person and target at 95% confidence level, assuming that both variances are unknown and unequal	• Hypothesis testing 2-sample • The mean of target is higher than mean of the maximum heart rate of the person
age and resting_bp	To determine the linear relationship between age of the person and blood pressure while resting of the person at 95% confidence level.	• Correlation • If the age of the person increase, the blood pressure while resting of the person also will increase that shows linear relationship
age and max_hr	To determine the relationship between age of the person (independent variable) and the maximum heart rate of the person (dependent variable)	• Regression • If the age of the person increase, the maximum heart rate of the person will also increase
target and sex	To test the relationship between target and sex of the person at 95% confidence level	• Chi-Square Test of independence • There is no relationship between the target and sex of the person

3.0) Data Analysis

3.1) Hypothesis Testing 2-sample

For hypothesis testing by 2 samples, we used maximum heart rate variable with disease and maximum heart rate without disease. First we are assuming that the variances of both variables are unknown and unequal. Then, find the degree of freedom by using formula.

Sample 1 : maximum heart rate of person with disease (target == 1)

Sample 2: maximum heart rate of person without disease (target == 0)

Tabulation data:

Values	
n1	165
n2	138
s1	19.17428
s2	22.59878
xbar1	158.4667
xbar2	139.1014

1. Hypothesis testing :

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 > \mu_2$$

Mean of the maximum heart rate of the person with disease is bigger than the maximum heart rate of the person without disease.

2. Test statistical value :

$$t_0 = \frac{x_1 - x_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$
$$t_0 = \frac{158.4667 - 139.1014 - 0}{\sqrt{\frac{19.17428^2}{165} + \frac{22.59878^2}{138}}}$$
$$= 7.9531$$

3. Degree of freedom :

$$v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{\left(\frac{s_1^2}{n_1} \right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2} \right)^2}{n_2 - 1}}$$

$$v = \frac{\left(\frac{19.17428^2}{165} + \frac{22.59878^2}{138} \right)^2}{\frac{\left(\frac{19.17428^2}{165} \right)^2}{165 - 1} + \frac{\left(\frac{22.59878^2}{138} \right)^2}{138 - 1}}$$

$$= 269.9026 \approx 269$$

These values get from R studio

With significance value = 0.05, the critical value is -1.6505

s2	22.59878
t.alpha	-1.65053787349266
t0	7.95307306045032
v	269.902570549481

4. Decision and Conclusion :

Test statistical value > critical value

$t_0 > t_{\alpha} : 7.9531 > 1.6505$

Thus, reject H_0 .

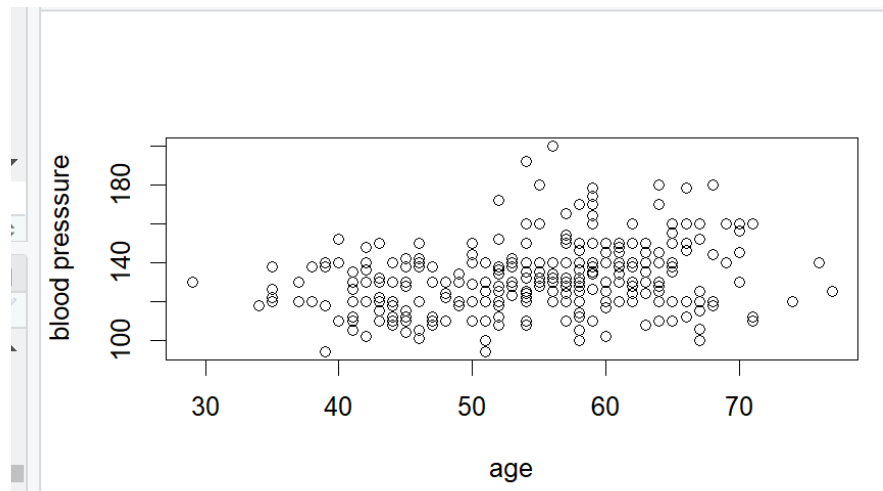
Therefore, there is no sufficient evidence to conclude that the mean of maximum heart rate of the person with disease is equal to the mean of maximum heart rate of the person without disease.

```
data: max_hr[target == "1"] and max_hr[target == "0"]
t = 7.953, df = 269.9, p-value = 5.019e-14
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 14.57132 24.15912
sample estimates:
mean of x mean of y
 158.4667  139.1014
```

3.2) Correlation

When we are using correlation analysis, we could determine whether the linear relationship is strong or weak by using Pearson's Product-Moment Correlation Coefficient. For this we used the age of the person variable and blood pressure while resting variable. We want to find the strength of the relationship between those two variables. We test the coefficient, r with the critical value.

1. Scatter plot



Scatter plot shows the relationship between age of the person and blood pressure while resting. The scatter plot shows that there is almost no correlation. So, supposedly we should get a correlation coefficient near to zero.

1. Sample correlation coefficient using Pearson's Product-Moment Correlation Coefficient

$$r = \frac{\sum xy - \left(\sum x \sum y \right) / n}{\sqrt{\left[\left(\sum x^2 \right) - \left(\frac{\sum x^2}{n} \right) \right] \left[\left(\sum y^2 \right) - \left(\frac{\sum y^2}{n} \right) \right]}}$$

By this formula in Rstudio, we get the r value is 0.2794. This is a positive correlation between age and blood pressure

2. Significance Test for Correlation

$H_0: \rho = 0$ (no linear correlation)

$H_1: \rho \neq 0$ (linear correlation exists)

$$t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$
$$t = \frac{0.2793509}{\sqrt{\frac{1-0.2793509^2}{304-2}}}$$
$$= 5.0559$$

t value is 5.0559 in Rstudio by using the formula above.

From the RStudio, we can get the p-value = 7.762e-07. From the t table, as we are using the two-tailed test with the degree of freedom = $n - 2 = 304 - 2 = 302$ and significance level $= 0.05 / 2 = 0.025$, the critical value, $t_{\alpha/2}$ calculated is ± 1.960 .

3. Decision and Conclusion :

t value is bigger than critical region : $t > \text{critical region}$

Reject the H_0 .

Thus, there is insufficient evidence to conclude that there is no linear relationship between age of the person and blood pressure while resting at a 95% confidence level.

```
Pearson's product-moment correlation  
data: CHD_data$age and CHD_data$resting_bp  
t = 5.0475, df = 301, p-value = 7.762e-07  
alternative hypothesis: true correlation is not equal to 0  
95 percent confidence interval:  
 0.1720897 0.3800657  
sample estimates:  
      cor  
0.2793509
```

3.3) Regression

For regression analysis, we used the age of the person variable and maximum heart rate of the person variable to test. We should find the relationship between both variables. Now, the independent variable is the age of the person and the dependent variable is the maximum heart rate of the person. Since the regression model is linear, hence we will use simple linear regression. We will investigate how changes in the age of the person could impact the maximum heart rate of the person. Assuming that variations in age of the person caused maximum heart rate of the person.

Mathematical Equation of Population Linear Regression:

$$y = \beta_0 + \beta_1 x + \varepsilon,$$

Where y is the dependent variable, β_0 (Population y-intercept), β_1 (Population slope coefficient) and x (Independent variable) is the linear component ε is the random error component.

1. Estimated Regression Model

$$\hat{y}_i = b_0 + b_1 x$$

I) Find the least-square criterion.

Find the value of b_0 and b_1 by using the formula:

a.
$$b_1 = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

b1	-0.607683177735747
n	304
x	int [1:303] 63 37 41 56 57 57 56 44 52 57 ...
y	int [1:303] 150 187 172 178 163 148 153 173 162 174 ...

Hence, from the Rstudio the value of b_1 is -0.6077

b.

$$b_0 = \bar{y} - b_1 \bar{x}$$

b0	182.684372893865
----	------------------

Hence from the RStudio the value of $b_0 = 182.6844$

II) Now we need to substitute both the value of b_0 and b_1 and apply it into the estimated regression model.

$$\hat{y} = 182.6844 - 0.6077x$$

Based on the formula above we can interpret that the coefficient of the intersection is denoted by b_0 and the coefficient of the slope is denoted by b_1 . Since there are no people with maximum heart rate of zero in the data, we can say that the age of the people that cannot be explained within the range of observed maximum heart rate of people is 182.6844. On the other hand, b_1 informs us that for every decreasing maximum heart rate of people, the age of the people will decrease by 0.6077.

III) Next, we need to do the Explained and Unexplained Variation.

$$SST \left(\sum (y - \bar{y})^2 \right) = SSE \left(\sum (y - \hat{y})^2 \right) + SSR \left(\sum (\hat{y} - \bar{y})^2 \right)$$

where $\bar{y}$ is the average value of the dependent variable
 y is the observed values of the dependent variable
 $\hat{y}$ is the estimated of y for the given x value

SSE	149244.354193192
SSR	9198.86032826034
SST	158443.214521452

From the RStudio we get the value for $SSE = 149244.3542$, $SSR = 9198.8603$ and $SST = 158443.2145$.

IV) So now, using the value of SSR and SST, we can find the Coefficient of Determination, R^2 .

$$R^2 = \frac{SSR}{SST}$$

<code>R2</code>	0.0580577739226243
-----------------	--------------------

From the RStudio the value of R^2 is 0.0581.

Thus, we can say that 5.81% of the variation maximum heart rate of a person is explained by variation in age of the people.

V) Next up, we need to find the Standard Error of Estimate by using the value we got for SSE.

$$s_e = \sqrt{\frac{SSE}{n - k - 1}}$$

<code>se</code>	22.2303082139838
-----------------	------------------

From the RStudio the value of $s_e = 22.2303$.

VI) We also need the Standard Deviation of the Regression Slope using s_e .

$$s_{b_1} = \frac{s_e}{\sqrt{\sum x^2 - \frac{(\sum x)^2}{n}}}$$

<code>sb1</code>	0.140849621992267
------------------	-------------------

From the RStudio we get the value of $s_{b_1} = 0.1408$.

2.t-test

I. Hypothesis Testing

$H_0: \beta_1 = 0$ (no linear relationship)

$H_1: \beta_1 \neq 0$ (linear relationship exists)

II. Test statistic

$$t = \frac{b_1 - \beta_1}{s_{b_1}}$$

t	-4.3144111367875
---	------------------

From the RStudio we get the value of $t = -4.3144$.

III. hence from the t table, we are using two-tailed test with the degree of freedom = $n-2 = 304-2=302$ and the significance level $= \frac{\alpha}{2} = \frac{0.05}{2} = 0.025$, the critical value is ± 1.960 .

IV. Conclusion

Since $t > t_{\alpha/2}$, we reject the H_0 . There is not sufficient evidence to conclude that the age of people will not affect the maximum heart rate of people at a 95% confidence level.

3. Perform Linear regression. We will use `lm()` function.

```
Call:
lm(formula = y ~ x)

Coefficients:
(Intercept)          x
    204.289       -1.005
```

We can read the values of the intercept coefficient and the value of the slope coefficient, x .

```
> summary(model)

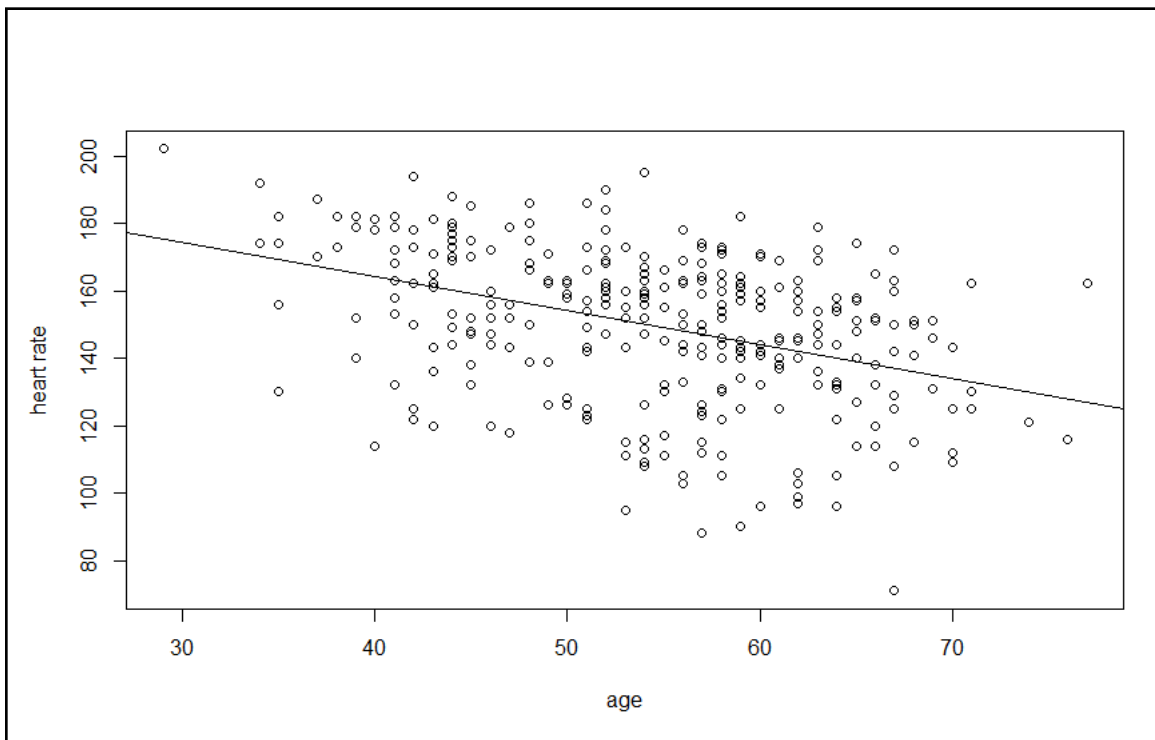
Call:
lm(formula = y ~ x)

Residuals:
    Min       1Q   Median       3Q      Max
-65.949 -11.954   3.975  15.921  44.985

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  204.2892     7.3485   27.800 < 2e-16 ***
x            -1.0051     0.1333   -7.539 5.63e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 21.04 on 301 degrees of freedom
Multiple R-squared:  0.1588,    Adjusted R-squared:  0.156
F-statistic: 56.83 on 1 and 301 DF,  p-value: 5.628e-13
```

From the summary, we can get the value of $b_0 = 204.2892$, $b_1 = -1.0051$, $s_{b1} = 0.1333$, $s_e = 22.2303$, degree of freedom = 302, $R^2 = 0.156$ and p-value = $5.628e-13$.



The scatter plot above is obtained by using the function `plot()` and the function `abline()`.

3.4) Chi-Square Test of independence

In our project 2, we are using the variables **target** and **sex** to show the relationship between the two qualitative variables. We expect that there exists a relationship between those two qualitative variables at 95% confidence level when we use a two way contingency table. The Rstudio below shows the observed frequency for sex and the target. The 0 & 1 in above shows the sex, male or female. 0 represents female and 1 represents male. The 0 & 1 on the left side shows the target, have disease or no disease. 0 represents no disease while 1 represents have disease.

	0	1
0	24	114
1	72	93

1) Hypothesis Statement :

H_0 : No relationship between the variables

H_1 : Variables have a relationship

2) Find the critical value :

$$\alpha = 0.05$$

$$df = (2-1)(2-1)$$

$$\text{Critical value, } \chi^2 = 3.8415$$

```
> x2.alpha  
[1] 3.841459
```

3) Calculate the expected counts :

Target	Sex				Total
	0		1		
	Obs	Exp	Obs	Exp	
0	24	$138 * 96 / 303 = 43.7228$	114	$138 * 207 / 303 = 94.2772$	138
1	72	$165 * 96 / 303 = 52.2772$	93	$165 * 207 / 303 = 112.7227$	165
Total	96	96	207	207	303

4) Calculate the test statistic value :

Cell, ij	Observed Count, o_{ij}	Expected Count, e_{ij}	$[o_{ij} - e_{ij}]^2 / e_{ij}$
1,1	24	43.7228	8.8967
1,2	114	94.2772	4.1260
2,1	72	52.2772	7.4409
2,2	93	112.7227	3.4508
$\chi^2 =$			23.9144

From Rstudio, we can get the test static value as below which is the same as our manual calculation in the above table. When we calculate the test statistic value by formula, we get the value for $\chi^2 = 23.9144$.


```
data:  tbl
X-squared = 23.914, df = 1, p-value = 1.007e-06
```

We get the test statistics value, $\chi^2 = 23.914$ with p-value = 1.007e-06 when we do the RStudio calculation.

5) State the decision :

Test statistic : $\chi^2 = 23.914$

Critical value : $\chi^2 = 3.8415$

Conclusion :

Since the test statistic value $>$ critical value, thus we reject the null hypothesis, H_0 at $\alpha = 0.05$. The test static value falls within the critical region, There is evidence of a relationship between the variables.

4.0) Conclusion

As we finish this report by analyzing the data for our dataset named *Survey on Heart Disease Prediction Among People*. Actually we learned a lot by collecting data and analyzing data by this project 2. Firstly, we get to know some links to get the existing dataset about this topic. For example, Kaggle, from kaggle we get the data set of this topic by Allena Venkata Sai Aby. so, by this we learned to collect data from it and export to excel for future use in R programming. In R programming also **we learned how to do programming to get scatter plots for correlation analysis and regression analysis . From the scatter plot we experienced to find the coefficient is positive or negative.**

Then, in the data analysis part we started with hypothesis testing by 2-sample. We used two variables named target and maximum heart rate. We did this test to identify the hypothesis between both means. We got mean of maximum heart rate of the person with disease is greater than maximum heart rate of the person without disease. By this we reject null hypothesis. After doing this test, we got to know about this testing by R programming.

Second one, we did correlation analysis on two variables named age of the person and blood pressure of the person while resting. By doing this test we were able to learn how to plot graph to this analysis in R programming. Finally, we got the graph for correlation analysis. We got r value in programming is 0.2794. So, if near zero means it was weak relationship and almost no correlation occurs. The graph was generated also same as the theory. Thus, we learned to do this by R programming.

Next, we did a regression analysis to determine the relationship between age of the person and maximum heart rate of the person. This analysis also we should generate graph in R programming. We learned this also to do calculations and generate graph for regression. Lastly, for the Chi-Square test of independence, we used two variables named target and sex. For this also we learned to do calculations in R programming. Then we calculate the observed frequency and expected frequency to do comparison for test statistical value and critical value.

In conclusion the **interesting part from four of us is, we were enjoying do R programming to get the graph for the correlation and regression analysis.** This is because generating the scatter plot is new to us in these chapters. Finally we were able to finish this project with a lot of knowledge.

5.0) Youtube link

Youtube link for presentation video of project 2 by group 1:

<https://youtu.be/ac3P49Vlnjw>

6.0) e-Portfolio Link

Navindran A/L Raghupathy :

<https://eportfolio.utm.my/view/view.php?t=Kjg6uXc7wUYC2iAaskLR>

Kuganes Varman A/L Balan :

<https://eportfolio.utm.my/view/view.php?t=OdZw5TutKfr4a1oXqFI7>

Kuganraj A/L Ramesh :

<https://eportfolio.utm.my/user/kuganraj-a-l-ramesh/seci1143-05-probability-statistical-data-analysis>

7.0) Reference

1. Benjamin, E. J., Muntner, P., Alonso, A., Bittencourt, M. S., Callaway, C. W., Carson, A. P., ... & American Heart Association Council on Epidemiology and Prevention Statistics Committee and Stroke Statistics Subcommittee. (2019). Heart disease and stroke statistics—2019 update: a report from the American Heart Association. *Circulation*, 139(10), e56-e528.
<https://www.ahajournals.org/doi/full/10.1161/CIR.0000000000000659>
2. Blue, G. M., Kirk, E. P., Sholler, G. F., Harvey, R. P., & Winlaw, D. S. (2012). Congenital heart disease: current knowledge about causes and inheritance. *The Medical Journal of Australia*, 197(3), 155-159.
<https://www.mja.com.au/journal/2012/197/3/congenital-heart-disease-current-knowledge-about-causes-and-inheritance>
3. Diedenhofen, B., & Musch, J. (2015). cocor: A comprehensive solution for the statistical comparison of correlations. *PloS one*, 10(4), e0121945.
<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0121945>